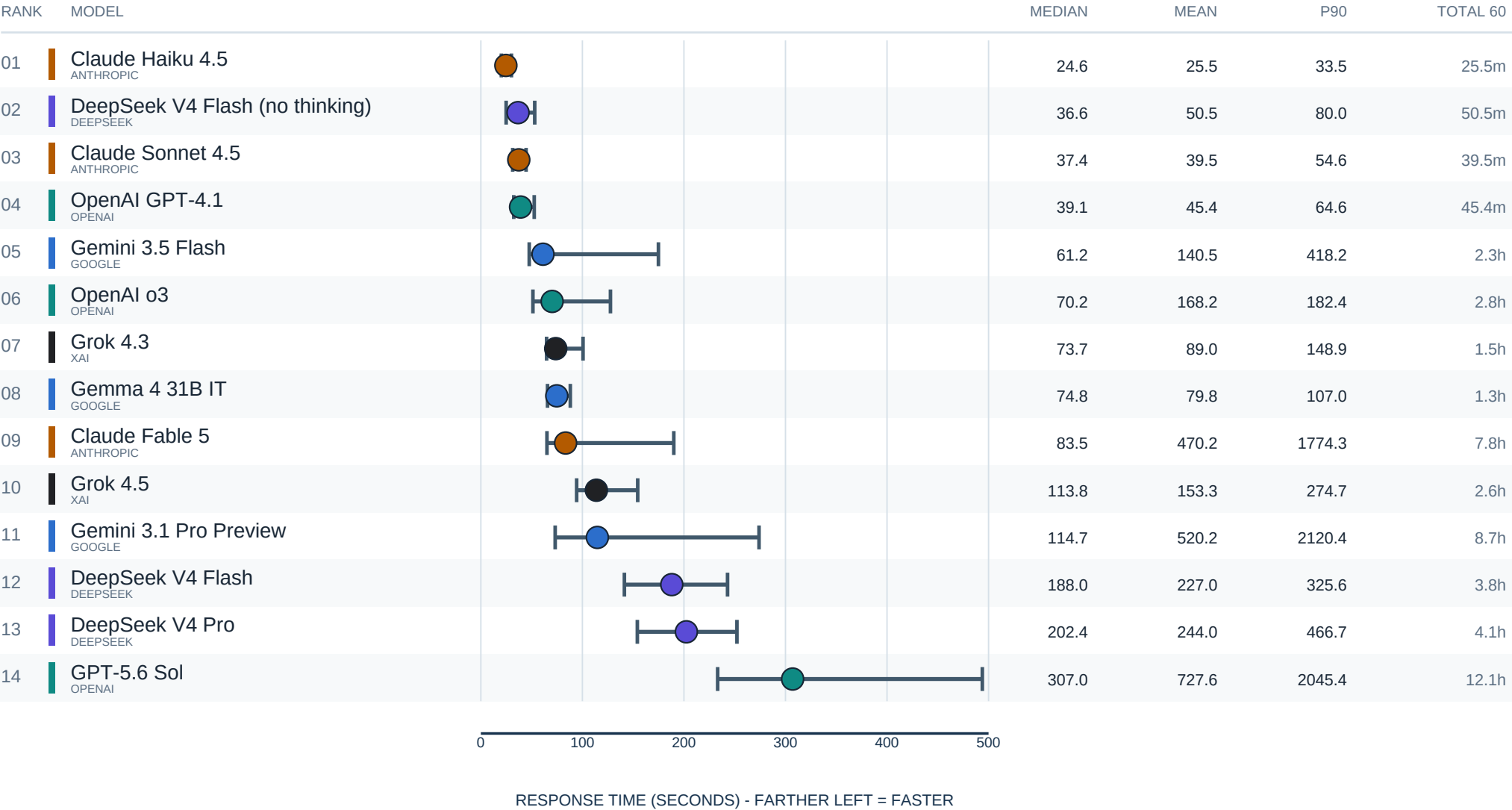


Typical response-time ranking

End-to-end time across 60 common exercises | ordered by increasing median

HOW TO READ  Dot = median; horizontal line with end caps = Q1-Q3 interval



Time excludes the local queue before dispatch; it includes the API call, generation, and any retries.

Where response times differ

Model-by-exercise heatmap | headers grouped by subject, exam session, and exercise



ANALYSIS 3		25 P1			25 P2			25 P3			25 P4			25 P5			26 P1A			26 P1B			26 P2A			26 P2B			26 P3			26 P4		
		E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3			
01	Claude Haiku 4.5	0.2	0.3	0.4	0.2	0.2	0.4	0.3	0.2	0.5	0.3	0.2	0.4	0.3	0.3	0.3	0.5	0.3	0.6	0.5	0.3	0.4	0.3	0.2	0.3	0.3	0.2	0.3	0.4	0.2	0.6	0.4	0.2	0.4
02	DeepSeek V4 Flash (no thinking)	0.3	0.4	0.4	0.5	0.8	0.5	0.5	2.4	0.7	0.6	0.9	0.3	0.3	0.5	0.9	0.6	0.7	0.9	0.5	0.6	0.8	0.5	1.3	0.4	0.6	0.5	0.4	0.6	0.7	0.4	0.4	0.3	0.6
03	Claude Sonnet 4.5	0.4	0.3	0.5	0.5	0.2	0.6	0.5	0.4	0.6	0.5	0.4	0.4	0.6	0.5	0.5	0.8	0.5	0.6	0.7	0.8	0.6	0.4	0.2	0.6	0.4	0.3	0.7	0.5	0.3	0.7	0.5	0.3	0.8
04	OpenAI GPT-4.1	0.4	0.3	0.3	0.5	0.3	0.5	1.3	0.4	0.5	0.7	0.5	0.5	0.5	0.6	0.5	0.8	1.0	0.8	0.9	0.4	0.6	0.4	0.4	0.6	0.4	0.3	0.5	0.6	0.2	1.9	0.7	0.2	1.0
05	Gemini 3.5 Flash	2.1	1.6	3.2	2.2	3.2	3.2	2.9	3.6	4.1	4.5	5.2	4.7	4.9	6.3	5.1	0.7	0.5	0.6	0.7	0.7	0.6	0.6	0.6	0.8	0.6	0.5	0.6	0.9	0.7	1.0	1.0	0.6	0.9
06	OpenAI o3	0.8	1.1	0.8	0.9	1.2	0.6	2.2	1.0	0.6	0.6	1.0	0.9	1.3	0.9	0.5	0.7	1.6	1.8	1.0	1.2	1.7	1.2	27.2	1.1	0.8	0.9	1.0	0.8	1.4	0.6	0.8	0.9	0.8
07	Grok 4.3	0.7	0.6	1.1	1.0	1.9	0.6	0.6	1.0	0.7	0.9	1.2	1.3	0.9	0.9	0.7	1.1	1.0	1.0	1.0	1.0	0.8	1.3	1.0	0.9	1.7	1.0	1.5	1.0	1.3	1.0	1.2	1.1	1.0
08	Gemma 4 31B IT	0.8	0.8	0.9	1.0	0.7	1.0	0.8	0.6	1.0	1.1	0.9	0.9	1.0	1.1	0.7	1.1	1.1	1.2	1.0	1.5	1.4	1.0	1.0	1.4	1.0	1.0	1.2	1.0	0.6	1.0	1.0	0.7	1.4
09	Claude Fable 5	2.5	4.1	4.7	1.0	0.8	1.4	0.9	1.0	1.0	1.1	1.0	1.2	1.0	0.9	1.1	1.0	0.9	1.0	1.0	1.0	1.1	1.0	1.0	1.1	1.0	1.1	1.0	16.6	10.0	1.3	21.9	4.1	16.6
10	Grok 4.5	1.2	0.9	0.9	1.3	1.6	1.2	1.0	3.0	1.1	1.3	2.7	1.3	1.4	1.3	1.5	1.4	1.4	1.8	1.3	1.3	1.6	1.3	3.1	4.4	2.8	2.1	1.9	1.3	1.7	1.0	1.2	1.7	1.3
11	Gemini 3.1 Pro Preview	4.0	2.2	2.0	1.0	0.8	1.0	1.0	1.0	1.2	0.9	0.9	0.9	0.8	1.1	10.9	1.0	0.8	0.9	0.8	0.9	0.9	0.9	0.7	0.9	24.9	20.1	0.9	3.4	0.6	1.7	0.8	1.5	6.6
12	DeepSeek V4 Flash	3.3	1.3	5.0	2.7	1.3	2.8	2.0	2.5	3.4	2.5	2.7	20.4	2.0	1.6	2.9	2.5	2.2	2.8	2.6	2.0	2.5	3.2	1.9	2.1	3.3	2.1	2.2	5.9	2.0	3.3	1.6	1.3	2.2
13	DeepSeek V4 Pro	2.2	1.9	2.4	1.8	3.4	2.1	2.0	2.3	2.5	1.6	3.8	1.1	1.9	2.8	2.7	3.0	3.0	2.7	3.0	2.1	3.0	2.4	2.7	3.1	3.8	2.8	2.1	3.4	5.6	3.2	2.5	1.3	0.9
14	GPT-5.6 Sol	8.1	4.6	7.1	2.2	2.0	4.8	2.5	2.8	4.1	26.2	2.7	3.3	3.4	2.3	4.3	3.4	2.8	3.6	3.5	3.0	3.2	3.6	2.1	1.8	3.1	2.1	1.8	13.1	6.0	6.9	19.4	15.3	48.6

PHYSICS 2			25 P1			25 P2			25 P3			25 P4			25 P5			26 P1			26 P2			26 P3			26 P4			
			E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3	
01		Claude Haiku 4.5	0.2	0.3	0.4	0.3	0.3	0.3	0.5	0.4	0.2	0.4	0.5	0.4	0.4	0.3	0.4	0.3	0.4	0.3	0.3	0.4	0.3	0.5	0.3	0.3	0.3	0.3		
02		DeepSeek V4 Flash (no thinking)	0.2	0.5	0.3	0.8	0.4	0.3	0.4	0.7	0.7	0.5	0.6	0.6	0.5	0.5	0.4	0.5	0.3	0.4	0.3	0.3	0.9	0.4	0.4	0.5	0.5	0.3	0.3	
03		Claude Sonnet 4.5	0.4	0.6	0.6	0.5	0.5	0.5	0.6	0.6	0.4	0.5	0.7	0.5	0.6	0.5	0.7	0.5	0.5	0.6	0.5	0.5	0.6	0.6	0.6	0.7	0.4	0.4	0.4	
04		OpenAI GPT-4.1	0.5	1.0	0.7	0.5	0.6	0.5	0.7	0.7	0.5	0.5	0.8	0.4	0.7	0.5	0.7	0.7	0.5	0.4	0.8	0.3	0.4	0.6	0.4	0.5	0.4	0.6	0.3	
05		Gemini 3.5 Flash	0.4	0.9	0.9	0.9	0.9	0.6	0.8	0.8	0.6	0.9	0.9	0.9	0.6	0.5	0.9	0.9	1.0	7.6	0.9	0.6	0.8	1.1	0.6	1.0	0.8	0.7	0.8	
06		OpenAI o3	1.6	0.9	0.7	0.5	0.6	0.9	0.9	0.7	2.4	0.7	1.0	0.9	3.9	0.9	0.6	2.3	0.9	0.9	1.7	1.6	1.4	0.7	1.1	1.4	1.1	0.8	0.7	
07		Grok 4.3	0.6	1.0	0.9	1.1	0.9	1.0	1.1	1.3	1.3	0.9	0.8	1.5	0.9	1.1	1.4	1.0	1.8	1.1	0.8	1.2	0.7	1.0	1.3	1.1	1.4	1.2	2.0	
08		Gemma 4 31B IT	0.7	1.0	1.1	0.9	1.1	1.0	1.2	1.1	0.7	1.2	1.3	0.9	1.1	1.5	1.1	1.1	1.0	0.8	0.9	0.8	0.9	1.0	0.9	1.1	0.9	1.3	0.7	
09		Claude Fable 5	35.5	35.1	54.7	22.7	42.7	52.9	0.9	0.9	0.6	1.1	1.0	1.1	0.9	0.7	0.9	0.9	0.9	0.9	13.9	1.1	0.7	1.1	0.8	0.9	0.9	0.8	0.6	1.2
10		Grok 4.5	1.3	1.9	1.3	1.6	2.5	1.9	1.1	2.1	3.0	1.8	1.5	2.2	1.8	1.6	1.9	1.0	3.5	0.9	1.7	1.9	3.8	1.1	1.9	1.0	1.4	2.2	6.6	
11		Gemini 3.1 Pro Preview	26.3	4.0	89.7	34.0	40.4	32.1	2.5	1.7	1.3	1.7	1.9	15.0	1.6	1.7	1.5	15.0	1.4	1.1	1.2	7.4	18.7	5.8	1.4	1.0	1.1	1.5	1.4	
12		DeepSeek V4 Flash	1.4	2.2	1.8	3.3	2.6	1.5	2.8	3.3	1.9	2.7	2.1	3.0	2.3	1.5	1.7	2.6	3.8	2.0	2.0	3.2	3.2	2.5	2.1	1.3	1.9	2.2	3.0	
13		DeepSeek V4 Pro	2.9	2.2	2.1	2.2	7.2	2.4	1.4	3.1	2.6	3.3	2.7	3.9	2.4	2.6	2.7	2.4	2.7	3.0	1.5	3.1	5.6	1.0	2.6	1.3	7.1	3.3	10.0	
14		GPT-5.6 Sol	34.6	32.3	75.6	31.0	69.0	43.2	3.7	5.3	3.6	4.9	2.5	5.9	3.5	2.2	4.0	4.1	2.6	4.3	3.9	3.7	4.1	3.3	4.0	4.6	3.2	2.6	5.3	

0.5 means half the exercise's typical time; 2.0 means twice as long. Normalization controls exercise difficulty.

Common panel: 60 exercises (33 Analysis 3, 27 Physics 2), 14 models, 840 macro-averaged cells

How mean response time is formed

Token volume and effective time intensity jointly explain observed response time



Seconds per 1K is an operational ratio: it includes latency, generation, and retries, not pure provider throughput.

Common panel: 60 exercises (33 Analysis 3, 27 Physics 2), 14 models, 840 macro-averaged cells

Response time vs accuracy

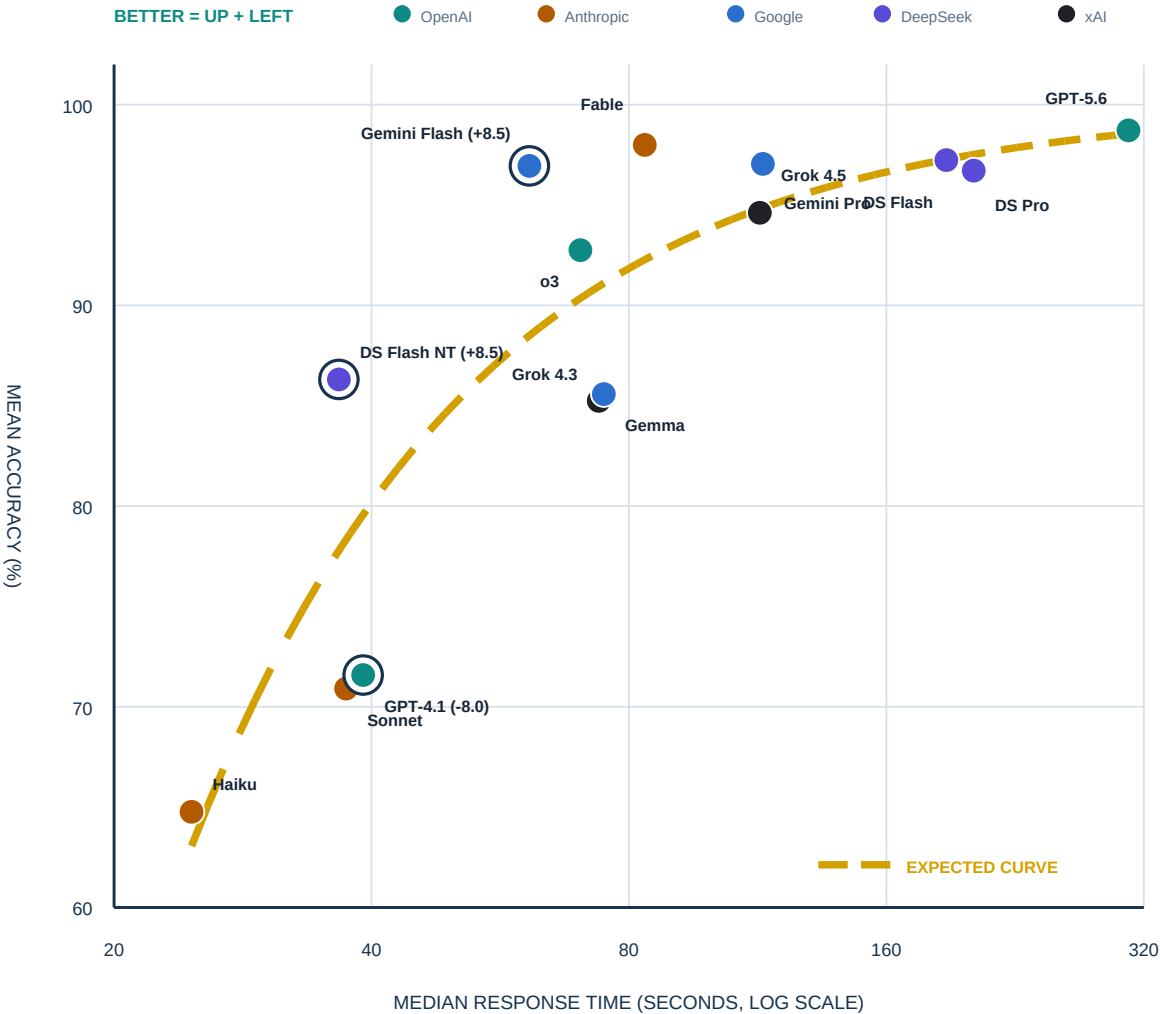
OPTION 4

Correlation and model deviations | 14 models x 60 common exercises

KEY RESULT Longer response time is associated with higher accuracy, but the gains flatten near the accuracy ceiling.

Pearson r (log time) = 0.80 | rank correlation ρ = 0.78 | bounded-curve R-squared = 0.78

1 OVERALL RELATIONSHIP



2 WHO BEATS OR MISSES THE TREND?

ACCURACY GAP (POINTS)

BELOW EXPECTED ABOVE EXPECTED



HOW TO READ: right of zero = more accurate than expected; left of zero = less accurate than expected. Rings mark the 3 largest gaps.